Task 1 - (20 points)

The most common type of therapy in health care is to prescribe medication to patients. Medication can be provided to patients in many different ways. It can come in the form of pills that have to be swallowed, fluid that the patient must drink, fluid that is injected, and so on. Because medication treatment is so widely spread it is also a main source for errors. Patients can get the wrong medication, a dose that is too high or medication that can't be used together with other medication they already use.

Because of its importance in the delivery of care, a big hospital decides to support the process of prescribing and giving medication (also known as 'medication management') by means of an information system. As you have learned, the first step in developing such a system is to get a thorough understanding of the medication management process as it is.

Based on a diagnosis of the patient's health problem, a doctor prescribes medication. A nurse is not allowed to do this. The prescription is written on a special form that is sent to the hospital's pharmacy. Here the prescription is checked by a pharmacist. If the prescribed medication does not fit other health problems the patient has or does not go together with medication already used, the prescription is cancelled, and the doctor is asked to revise it. If it is ok, the pharmacy prepares the medication. If the patient is admitted to the hospital, it is sent to the department where the patient is hospitalized. If the patient is not in the hospital it is sent to the pharmacy's counter where it can be collected by the patient.

The medication is given to or taken by the patient according to the prescription. This is called administering the medication. For example, the patient swallows a pill three times a day for a week. Or in the hospital a fluid is injected by a nurse every day for five days in a row. After the medication has been taken (note that this can take several days or even weeks), the patient's health status is checked to see if the treatment works. This is done by the doctor. If the treatment seems to work but the health problem has not disappeared completely, the treatment will be prolonged. This basically means that a new prescription is issued. If the health problem has disappeared, the treatment is finished. If it proves that the treatment does not work, it is also finished. The health problem of the patient must be assessed again because the diagnosis was apparently wrong.

The model given below contains several mistakes, both syntactical and semantical mistakes. Find at least three syntactical and three semantical mistakes (so, six mistakes in total). Motivate why you consider these to be mistakes.

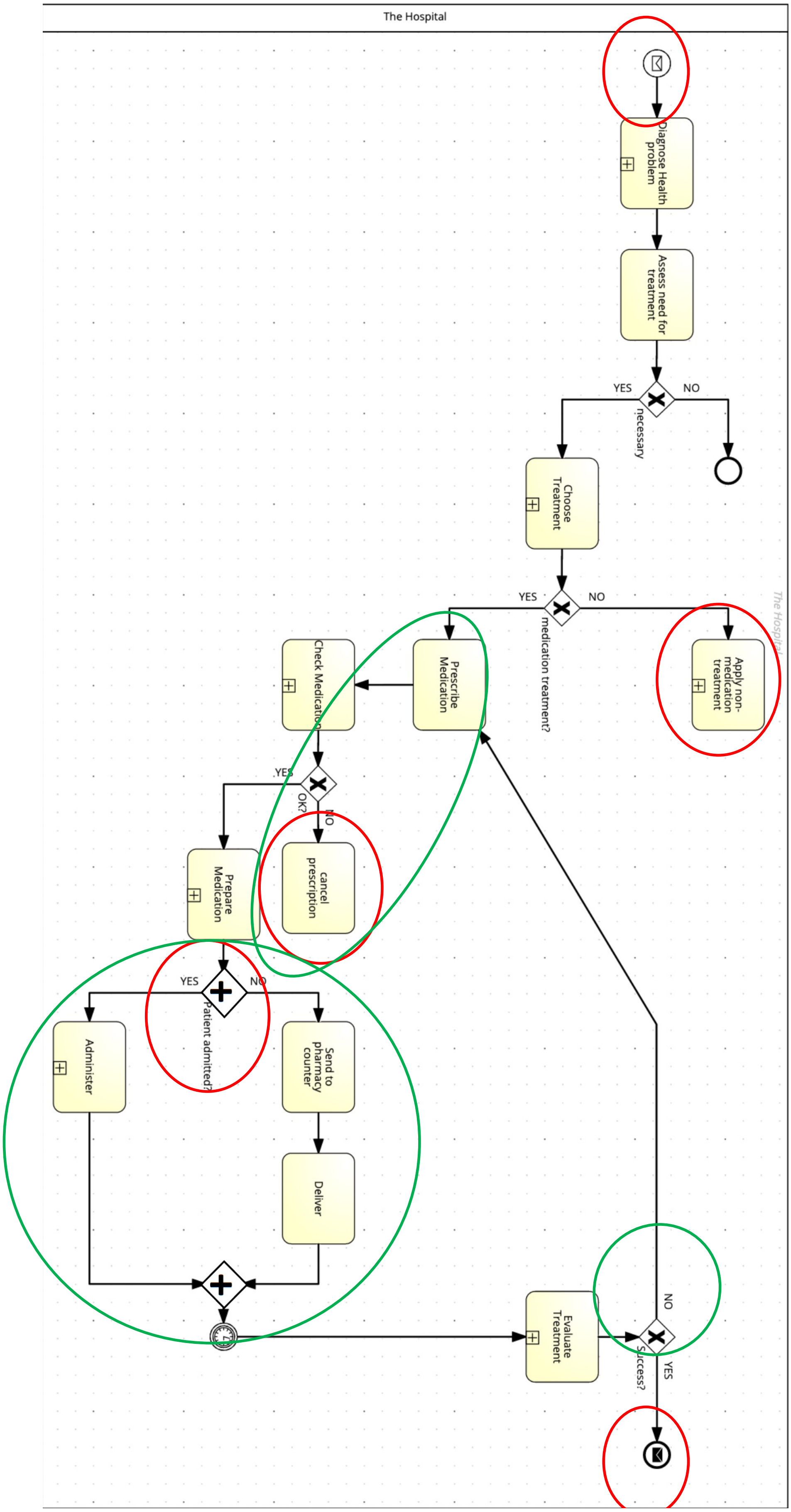
The red circles in the figure below indicate syntactical mistakes:

- Both the start and the end event are modelled as message events but there is no message flow connected
- The two activities are dangling, no outgoing flow
- The AND-gateway can't have a condition and a 'Yes'/'No' outgoing pair

The green circles indicate semantical mistakes:

- The text indicates that in case the prescription is cancelled it must be redone. So, there should be an arrow from 'cancel prescription' to 'prescribe medication'
- It is wrong to model the part after 'prepare medication' as an AND split. It should be an XOR split
- If the treatment is not successful, two things can happen. The treatment can be prolonged, so a new round of the medication is prescribed (correctly modelled by the arrow) or it is decided that the diagnose was incorrect and the patient must be diagnosed again. So, we miss a decision activity, a XOR split and an arrow back to 'Diagnose patient'.
- It could also be remarked that no lanes where used. Not really a semantical mistake, but loss of important information for the system development

3 points per correctly identified mistake, 2 points for correct use of semantical vs syntactical

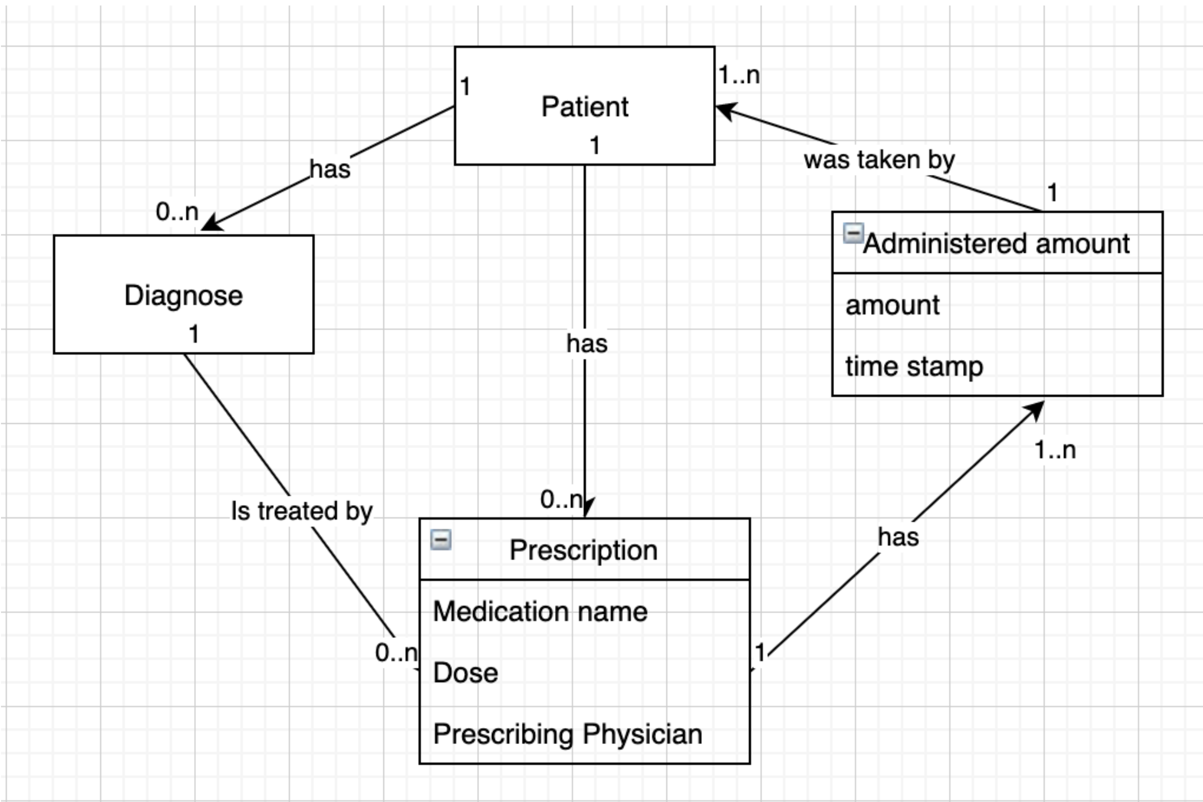

The Hospital
Diagnose Health problem
Assess need for treatment
YES
NO
necessary
Choose Treatment
YES
NO
medication treatment?
Apply non-medication treatment
Prescribe Medication
Check Medication
YES
NO
OK?
cancel prescription
Prepare Medication
YES
NO
Patient admitted?
Administer
Send to pharmacy counter
Deliver
Evaluate Treatment
NO
YES
Success?

Task 2 (5 points)

What is a computer-based information system (CBIS)?

Computer-based information system is a single set of hardware, software, databases, telecommunications, people, and procedures that are configured to collect, manipulate, store, and process data into information.

Task 3 (15 points)



The simple data model above represents the information that will be at the core of the medication management system that will be developed and implemented. In the first lecture and in chapter 1 of the book, quality of information was discussed. A table was presented with 11 characteristics of information quality. Discuss at least three of these characteristics that you consider to be crucial for the success of the medication management system.

This is the table referred to:

Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. (This is commonly called garbage in, garbage out [GIGO].)
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory.
Relevant	Relevant information is important to the decision maker. Information showing that lumber prices might drop might not be relevant to a computer chip manufacturer.
Reliable	Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up might not be reliable.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not overly complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

There are many possibilities here, although I think it will be difficult to argue for the importance of 'Economical' and 'Flexible'. The students get:

- 5 points for coming up with three factors from the table (they don't have to use the exact same name)
- 5 points for demonstrating an understanding what these characteristics mean
- 5 points for a sound argumentation why exactly these characteristics are important for the case at hand

Task 4 (5 points)

What is a decision support system (DSS)?

Decision Support system is an organized collection of people, procedures, software, databases, and devices used to help make decisions that solve problems. These problems can be semi-structured or unstructured.

Task 5 (15 points)

One of the main goals with the development and implementation of the medication management system is to increase patient safety. That includes that a patient gets the right diagnose as soon as possible, the right treatment and that mistakes are reduced to a minimum. One way to achieve this is to add ***decision support system (DSS)*** functionality to the system. Which parts of the process presented in Task 1 would you support by DSS functionality? Describe briefly how this functionality supports the decision making. Note that DSS functionality could be useful at several places in the process.

Candidates are:

- Diagnose health Problem
- Choose Treatment
- Check Medication
- Evaluate treatment

In the description the students can take different routes. They can for example focus on the functionality that is offered to support the problem solving, or on the components the system is built of. The most important is that they show an understanding that an DSS supports decision making by humans, they don't fully automate. So, they should give users input on the basis of which decisions are made in the context of solving a problem (such as: what is the health problem of this patient, or which treatment is applicable?)

Task 6 (10 points)

How would you characterize the introduction of the medication management system: as an example of ***reengineering*** or ***continuous improvement***? Motivate your answer.

The students should in their answer explain what these concepts mean. The question is deliberately a bit unspecific. The right answer is dependent on what they understand this medication management system to be. If it is a modestly advanced tool that supports the activities and actors described in the case for Task 1, it is continuous improvement. However, if you let your creativity run loose and imagine a system with AI/DSS components, and maybe an automatic medicine dispenser, it could actually result in a quite different process. Roles of actors change or even disappear, activities change or are removed. In that case it would be more reengineering.

Task 7 (10 points)

As part of the hospital's quality management program information on unwanted deviations or mistakes is collected from the medication management system. Examples of this information are:

1. Patients being diagnosed wrongly
2. The wrong treatment given a diagnose
3. Medication was prescribed that could not be combined with medication the patient already had
4. The wrong medication given to a patient
5. The wrong amount of medication given to a patient

The collected information is used in an information system (system X) that can aggregate, combine and visualize it. The book and the lectures use a typology of information systems that includes: Transaction Processing Systems; Customer Relationship Management Systems; Supply Chain Management Systems; Decision Support Systems; Management

Information Systems; Executive Support Systems. How would you classify system X? Motivate your answer.

The list of system types is not exhaustive, therefor the word ‘inlcudes’ is used. There are other types of systems referred to in the book and/or the lectures, such as ERP systems, knowledge management systems. The right answer here is either MIS or ERP.